

ADITI TIWARI

aditit5@illinois.edu | <https://adititiwari19.github.io/> | <https://www.linkedin.com/in/adititiwari2000>

RESEARCH INTERESTS

I study multimodal world models that build persistent representations of dynamic environments for reasoning, forecasting, and control. My work focuses on learning and updating structured world state from multimodal observations to support prediction, causal intervention, and long-horizon decision-making.

EDUCATION

University of Illinois, Urbana-Champaign (UIUC)

PhD in Computer Science

Urbana, IL

Aug 2024-Present

- CGPA: **3.9/4.0**
- Advisor: **Dr. Heng Ji**
- Fellowships/Grants:
 - Amazon Alexa Research Gift (Fall 2026). Authored the research proposal selected for funding by Amazon.
 - Adobe Research Grant (Spring 2026)
 - UIUC Computer Science Departmental Fellowship (2025-2026)
 - Thomas M. Siebel School of Computing and Data Science Fellowship (2024-2025)

University of Illinois, Urbana-Champaign (UIUC)

Master of Science (MS) in Computer Science

Urbana, IL

Aug 2022-May 2024

- CGPA: **3.85/4.0** | **Courses:** Computer Vision, Computational Photography, Autonomous Vehicle System Engineering, Applied Machine Learning, Applied Parallel Programming
- **Thesis:** Action Detection in In-Classroom Firefighter Training: A 360-Degree Video Analytics Service
- Advisor: **Dr. Klara Nahrstedt**
- MS funded by the **National Science Foundation (NSF)** grant number NSF IIS 21-40645

Guru Gobind Singh Indraprastha University

Bachelor of Technology in Computer Science

Delhi, India

Aug 2018-Jul 2022

- CGPA: **9.39/10.0** | **Gold medalist** | **Courses:** Data Structures, Database Systems, Operating Systems, Computer Networks

HONORS AND AWARDS

- ICML Gold Reviewer Award, 2026
- Spotlight Recognition, NeurIPS 2025, Datasets and Benchmarks Track, Fire360
- Best Paper Award, IEEE International Conference on Smart Computing (SMARTCOMP), ACT360, 2025
- Best Poster Award (Research Mentor), Illinois Undergraduate Research Scholars (URSA) Symposium, UIUC, 2025. Mentored an award-winning project on counterfactual reasoning in video-language models.
- Recognized on the List of Teachers Ranked as Excellent by Their Students for CS107: Introduction to Data Science (Fall 2023, Winter 2023–2024)
- GNIT Fellowship Award for Undergraduate Research, 2022
- Best Research Poster Award, GN Institute of Technology Undergraduate Research Presentation Competition, 2022
- Gold Medalist, Bachelor of Technology in Computer Science, GNIT, Guru Gobind Singh Indraprastha University, 2022.
- Third Place, Robotryst Championship, Qualified for Nationals, IIT Delhi, 2016

PUBLICATIONS

Peer-Reviewed Conference and Journal Papers

2026:

- Feature Recovery for Object Understanding After Irreversible Physical Damage. NeurIPS 2026.
- Advancing Creative Physical Intelligence in Large Multimodal Models. NeurIPS 2026.
- ViTAL-X: Video-Text Alignment with Cross-Modal Temporal Edits. ECCV 2026.
- SceneDiff: Geometric-Semantic Consistency for Multiview Change Detection. ECCV 2026.
- CreativityBench: Evaluating Agent Creative Reasoning via Affordance-Based Tool Repurposing. COLM 2026.

2025:

- Fire360: A Benchmark for Robust Perception and Episodic Memory in Degraded 360-Degree Firefighting Videos. NeurIPS 2025, Datasets and Benchmarks Track. **Spotlight.**
- Model Context Protocol for Vision Systems: Audit, Security, and Protocol Extensions. NeurIPS 2025, LAW and Scaling Environments for Agents (SEA) Workshops. Also published in Springer, Communications in Computer and Information Science (CCIS), vol. 2759.
- MAGNET: Augmenting Generative Decoders with Representation Learning and Infilling Capabilities. ACL 2025, Main Conference.

- ACT360: An Efficient 360-Degree Action Detection and Summarization Framework for Mission-Critical Training and Debriefing. IEEE SMARTCOMP 2025. [Best Paper Award](#).

2024:

- ST-360: Spatial-Temporal Filtering-Based Low-Latency 360-Degree Video Analytics Framework. ACM Transactions on Multimedia Computing, Communications and Applications, 2024.

2023:

- Latency-Aware 360-Degree Video Analytics Framework for First Responders Situational Awareness. ACM NOSSDAV 2023.

Workshop Papers and Refereed Proceedings

- CausalSliders: Graph-Guided LoRA Interventions for Causally Consistent Image Editing. ICLR 2026, World Models and ReALM-GEN Workshops.
- CAMCUE: Predicting Camera Pose from Perspective Descriptions for Spatial Reasoning. ICLR 2026, World Models Workshop.
- Stress Tests REVEAL Fragile Temporal and Visual Grounding in Video-Language Models. ICML 2026, Combining Theory and Benchmarks Workshop.

Preprints

- Does Video Memory Use What It Retrieves? A Causal Audit of Memory Specificity.
- Competence-Gated Pooling of Language Models and Priors for Event Forecasting.
- OSExpert: Computer-Use Agents Learning Professional Skills via Exploration.

Master's Thesis:

- *Action Detection in In-Classroom Firefighter Training: A 360-Degree Video Analytics Service*. University of Illinois Urbana-Champaign, 2024.

RESEARCH EXPERIENCE

Adobe Research

Research Scientist Intern

San Jose, CA

May 2026-Present

- Developed a causal evaluation framework to distinguish genuine episodic memory use from generic feature support in video models. Introduced read-time memory replacement and corruption interventions to identify which retrieved evidence causally affects predictions.
- Found that some world-model benchmarks preserve much of their measured memory benefit without episode-specific content, while tracking models depend more strongly on the correct stored evidence.

Confidential Research Project (Stealth)

Research Intern (Molecular Optimization & Drug Discovery)

Remote

Jan 2026-May 2026

- Designed and developed machine learning models for molecular optimization using real-world experimental datasets (ADMET, bioactivity), focusing on constraint-aware generation and structure-property modeling.

Adobe Research

Research Collaborator - Counterfactual Image Editing

Remote

Sept 2025-Jan 2026

- Designed and implemented *CausalSliders*, a parameter-space framework for counterfactual image editing that learns low-rank LoRA adapters for distinct causal factors (age, smile, gender) and composes them via a directed causal graph.
- Developed a custom training pipeline with intervention, minimality, and conditional-independence losses, achieving >85% causal accuracy and <10% attribute leakage on CelebA while preserving image quality (FID < 25, LPIPS < 0.25).
- Built a gating-based composition engine enabling real-time inference (~0.05s per image) and delivering ~1000× speedup over optimization-based baselines such as Deep-SCM, with strong generalization and fairness across facial and synthetic domains.

HERE Technologies

Research Collaborator - GeoDynamics: GPS-Satellite Trajectory Forecasting World Model

Remote

Nov 2025-Present

- Developing a multimodal foundation model that jointly embeds satellite imagery and GPS trajectory time-series for geospatial understanding.
- Building pixel-aligned embeddings and segmentation masks that align real GPS traces with visual context for road network localization, geo-tagging, and urban decision-support simulation.

Adobe Research

Research Scientist Intern

San Jose, CA

May 2025-August 2025

- Developed an end-to-end Model Context Protocol (MCP) server to connect LLM agents with Adobe's internal models and tools, enabling AI-driven workflows through agentic tool orchestration and semantic input-output alignment.
- Led a vision-focused study of 91+ MCP servers, producing a survey paper that identified protocol-level gaps in schema design, tool planning, and secure execution.
- Paper accepted to the NeurIPS 2025 Workshop on Bridging Language, Agent, and World Models for Reasoning and Planning.

Adobe Research

Research Collaborator - MAGNET Project

Remote

Dec 2024-Feb 2025

- Co-developed MAGNET, a unified training framework enabling decoder-only LLMs to perform both bidirectional representation learning and text infilling via a hybrid attention mechanism.
- Contributed to developing evaluation pipelines for benchmarking MAGNET-enhanced LLMs on tasks like MMLU, showcasing gains in representation robustness and contextual infilling.
- Adapted MAGNET training to the SlimPajama dataset using multi-objective self-supervised loss functions, and integrated with HELM for systematic evaluation on representation learning and reasoning benchmarks.

Rivian Automotive

Research Intern (Computer Vision)

Palo Alto, CA

May 2024-Aug 2024

- Designed and coded custom Image Signal Processing (ISP) pipeline from scratch to convert raw sensor data into RGB/YUV formats for human evaluation. Used low-level language to ensure performance efficiency, scalability, and maintainability.
- Implemented core functions like histogram equalization, demosaicing, decompanding, gamma correction, etc. in C/C++, achieving end-to-end latency of 2-4 seconds, surpassing company benchmarks, and reducing manual input by 85%.

Coordinated Science Laboratory, UIUC

Research Assistant (Computer Vision/Machine Learning)

Urbana, IL

Sep 2022-May 2024

- Curated one of the first action detection datasets of 360-degree firefighting videos and developed a custom annotation script.
- Developed ACT360, a quantized, latency-efficient action detection model for 360-degree videos, surpassing baseline YOWO with 4% mAP improvement (0.865) while achieving 48ms/frame inference.
- Optimized ACT360 for deployment, implementing NMS, confidence threshold tuning, and quantization-based compression, reducing inference time to 33ms while maintaining 0.850 mAP, enabling real-world firefighter training applications.
- ACT360 made video analysis during firefighter training 95% faster. Model successfully deployed at Illinois State Fire Department to be used during training sessions of firefighters.

Indian Institute of Technology (IIT)

Machine Learning Research Intern

Delhi, India

May 2021-Sep 2021

- Engineered UAV for autonomous object detection and route adaptation, attaining accuracy of **88.21%** (matching SOTA).

INDUSTRY EXPERIENCE

McAfee Inc.

Software Engineer Intern

Bangalore, India

Aug 2021-Dec 2021

- Automated HAR file processing and SQL-based data extraction using Python and Bash, reducing processing time by 58% and saving 6 person-hours per week. Also developed a customer query interface and analytics dashboard using AngularJS and JavaScript.

MEDSupervision (Startup)

Full Stack Developer Intern

Delhi, India

Nov 2020-May 2021

- Developed a telehealth platform with Twilio-based video calling, redesigned patient and admin portals, and optimized a database of over 10,000 records, increasing user engagement by 37% and improving data retrieval speed by 50%.

QikCircle (Startup)

Software Development Engineer Intern

Delhi, India

Feb 2020-May 2020

- Developed full-stack web features using JavaScript, Python, Flask, and SQL, increasing user engagement by 25% and reducing account creation time by 40% through database optimization.

MENTORING AND TEACHING

Mentoring:

- **Research Mentor, UIUC** (Aug 2023–Present)
Mentored 5 students across UIUC, UMD, and NTU Singapore on research projects in world models, video understanding, and multimodal reasoning.
 - Sofia Stoica, Haengbok Chung, Aashrith Bandaru (UIUC)
 - Krish Patel (University of Maryland)
 - Zihang Lin (Nanyang Technological University, Singapore) (intern in BLENDER Lab Summer 2026)
- **Illinois Undergraduate Research Scholars (URSA) Mentor** (Aug 2025-Dec 2025)
 - Undergraduate Research Mentees: Tia Atul Shrikhande; Abhay Pokhriyal
 - Best Research Poster Award
 - Project Goal: Our project explored counterfactual world models for video understanding, focusing on how AI systems can imagine alternative outcomes in instructional video clips (e.g., “what if the tool were not used?”). Guided the students in dataset creation, running pretrained vision-language models (CLIP, Whisper, VideoMAE), and designing lightweight evaluation metrics for counterfactual reasoning.

- **Undergraduate Thesis Co-advisor, UIUC**

(Aug 2023-May 2024)

- Student: [Haozhen Zheng](#) (Senior Thesis), co-advised with Prof. Klara Nahrstedt
- Our research centered on real-time detection of actions in 360-degree videos, employing advanced image segmentation models to enhance the precision and efficiency of the detection process.

Teaching:

- **Graduate Teaching Assistant, UIUC**

Urbana, IL

- *CS441 Applied Machine Learning* Aug 2025-Dec 2025
- Teaching assistant for a class with over 500 students. Led weekly discussion sessions and office hours, guided students through applied machine learning assignments, and provided individualized support. Contributed to course infrastructure and grading, and facilitated engagement in a large-scale learning environment.

- **Graduate Teaching Assistant, UIUC**

Urbana, IL

- *CS107 Data Science Discovery* Aug 2023-Dec 2023
- Ranked on the [list of excellent teachers](#) by students
- Lab instructor for a class with over 1000 students, where I spearheaded lab sections, delivered lectures on data science concepts, contributed to course content development, including guides for intricate data science topics, and conducted regular office hours to provide additional support and clarification.

SKILLS

- **Programming:** Python, C/C++, Bash, SQL
- **ML Frameworks & Tools:** PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, NumPy, Weights & Biases
- **Research Areas:** Multimodal Foundation Models, Vision-Language Models, World Models, Video Understanding, Computer Vision, Causal Representation Learning, Model Evaluation and Robustness
- **Model Training & Data:** Multi-GPU Training, Distributed Training, Parameter-Efficient Fine-Tuning (LoRA), Post-Training, Large-Scale Multimodal Data Curation, Synthetic Data Generation
- **ML Systems & Optimization:** CUDA, Triton, GPU Profiling, CUDA Kernel Optimization, Quantization, Model Compression, Scalable Training Infrastructure
- **Infrastructure & Development:** AWS S3, Docker, Git, Linux, JupyterLab

ACADEMIC SERVICE

- **Conference Reviewer:** ICLR (2026, 2027), NeurIPS (2026), ICML (2026, **Gold Reviewer Award**), AAAI (2027), WACV (2027), NLDL (2027), TMLR (2026)
- **Workshop Reviewer:** NeurIPS 2026 PhysUnderstand; NeurIPS 2025 workshops on [LAW: Bridging Language, Agent, and World Models](#), [CauScien: Uncovering Causality in Science](#), and [AI for Science](#)
- **Artifact Evaluation Committee:** [ACM SIGCOMM 2025](#)
- **Organizer:** UIUC IMMERSE Seminar Series (Fall 2024-Spring 2026)
- **Organizer:** UIUC Vision Lunch Seminar Series (Fall 2025-Fall 2026)
- **Organizer:** UIUC NLP External Speaker Series (Fall 2026)

MEDIA COVERAGE

- **"Illinois AI research among the top five most-published at NeurIPS 2025,"** Siebel School of Computing and Data Science, Jan 2026. Featured *Fire360* and *Model Context Protocol for Vision Agents*. [link](#)
- **"Illinois AI research spotlighted at Micro-NeurIPS and NeurIPS 2025,"** Illinois ECE, Dec 2025. Featured *Fire360* and *Model Context Protocol for Vision Agents*. [link](#)
- **Best Paper Award at IEEE SMARTCOMP 2025,** Siebel School announcement, 2025. [LinkedIn](#) · [Instagram](#)

LEADERSHIP, OUTREACH AND VOLUNTEERING

- Graduate Student Council, IMMERSE Seminar, UIUC (2025–Present)
- Intern Community Captain, Adobe Research (2025)
- Graduate Student Ambassador (PhD), UIUC (2025)
- Rivian Volunteer Program Participant (2024)
- Chapter Lead, Anita.Org (2021)
- Brand Ambassador, National Digital Literacy Mission, Women International Network (2021)
- C++ Mentor, GirlScript Foundation, Uplift Project (2019–2020)
- Member, Women Who Code, Delhi (2019–Present)
- Founder and President, National Service Scheme (NSS), GNIT, GGSIPU (2019)
- Head of Sports Society, GNIT, GGSIPU (2018)
- Board Member, Women International Network, Delhi (2018–Present)